Question	Answer	Marks	Partial Marks	Guidance
4(a)	135	**2**	**M1** for $\frac{360}{5+3} \times k$, where $k = 1$, 3 or 5 oe	M1 implied by 45 or 225 Alt method e.g. $w : 360 - w = 3 : 5$ $\frac{w}{360-w} = \frac{3}{5}$ Award M1 here if answer incorrect $5w = 1080 - 3w$ $8w = 1080$ $w = 135$
4(b)(i)	26	**2**	**B1** for $\angle ABD = 49$	$\angle ABD$ may be on diagram.
4(b)(ii)(a)	86	**2**	**B1** for $\angle QAD = 49$ or for $\angle BDA = 45$ or for $\angle BCA = 45$	For B1 angles must be identified or marked on diagram
4(b)(ii)(b)	Angle in a semicircle = 90	**1**		Must state angle in semicircle and 90 or angle subtended from diameter and 90 Accept angle at centre is twice angle at circumference and 90 (must state angle, circumference, centre and either twice oe or half oe if reversed) See AG
4(c)(i)	$[2 \times]\cos^{-1}\left(\frac{6.75}{11.5}\right)$ oe	**M2**	**M1** for $\cos(\ldots) = \frac{6.75}{11.5}$ oe	May see longer methods e.g. M2 for $[2\times]\left(180 - 90 - \sin^{-1}\left(\frac{6.75}{11.5}\right)\right)$ If other methods used $AT = 9.31$ or 9.310 to 9.311
	108.117…	**A1**		A1 after M1 implies M2

Question	Answer	Marks	Partial Marks	Guidance
4(c)(ii)	12.7 or 12.73 to 12.74	**2**	**M1** for $\frac{108.12}{360} \times 2 \times \pi \times 6.75$	Allow M1 for use of 3.14 or $\frac{22}{7}$ for π Allow e.g. 108, 108.1, 108.117…. for the angle for M1 M1 implied by answer 12.74…
4(c)(iii)	100 or 100.1 to 100.2	**3**	**M2** for $\frac{360-108.12}{360} \times \pi \times 6.75^2$ oe or **M1** for $\frac{108.12}{360} \times \pi \times 6.75^2$ If 0 scored, **SC1** for $\frac{360-108.12}{360} \times \pi \times k$	For method marks allow 3.14 or $\frac{22}{7}$ for π Allow e.g. 108, 108.1, 108.117…. for the angle for M2, M1 or SC1 M2 e.g. $\pi \times 6.75^2 - \frac{108.12}{360} \times \pi \times 6.75^2$ Implied by 43[.0] or 42.96 to 43.01 $k > 0$